

ENPH 334/PHYS 334 Electronics for Applied Scientists and Physicists

Course Syllabus – Fall Term 2026

Land Acknowledgement

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory. See: <http://www.queensu.ca/encyclopedia/t/traditional-territories>

Inclusivity Statement

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: <https://www.queensu.ca/inclusive/>

Course Overview

The objective of ENPH 334/PHYS 334 is to give students an introduction to basic analog and digital electronics as a prelude to further courses in the general electronics area, to courses (labs and design projects) using applied electronics, or to applications in your future careers as scientists or engineers.

This course is divided into two parts: analog electronics and digital electronics. The analog section deals with applied electronics, with an emphasis on operational amplifier (op-amps) based circuits. The digital logic section deals with logic gates and digital systems.

Following a brief review of basic components, we will study op-amp circuits and the effects of feedback to produce different mathematical operations and discuss applications of op-amps as amplifiers and filters. The digital logic section starts with binary arithmetic and Boolean algebra. It then continues with gates, latches, flip-flops, counters, shift registers, digital-to-analog converters (DACs), and analog-to-digital converters (ADCs). Finally, we will briefly examine memory used in electronic circuits and computers.

Instructor

Prof. Bhavin J. Shastri, Stirling 308F (bhavin.shastri@queensu.ca)

Teaching Assistants

There are five teaching assistants (TAs) who will be the primary contact for the labs and quizzes:

- Fraser McCauley, Head TA (fraser.mccauley@queensu.ca) – Labs
- Sam Lamontagne (gqb5@queensu.ca) – Labs
- Ammar Ibrahim (bv56@queensu.ca) – Labs
- Cameron Ingo (23vqp@queensu.ca) – Labs
- Arpan Sur (bv56@queensu.ca) – Quizzes/Labs

Calendar Descriptions

ENPH 334	Electronics for Applied Scientists	F 3–1.5–0.5; 5.0
The design of electronic circuits and systems, using commonly available devices and integrated circuits. The properties of linear circuits are discussed with reference to the applications of feedback; operational amplifiers are introduced as fundamental building blocks. Digital circuits are examined, and the properties of the commonly available I.C. types are studied; their use in measurement, control and signal		

analysis is outlined. Laboratory work is closely linked with lectures and provides practical experience of the subjects covered in lectures. (0/0/0/27/27)	
Prerequisites: ELEC 221	Exclusions: ENPH 333 (PHYS 333)

PHYS 334/3.0	Electronics for Physicists	
The design of electronic circuits and systems, using commonly available devices and integrated circuits. The properties of linear circuits are discussed with reference to the applications of feedback; operational amplifiers are introduced as fundamental building blocks. Digital circuits are examined, and the properties of the commonly available I.C. types are studied; their use in measurement, control and signal analysis is outlined. Laboratory work is closely linked with lectures and provides practical experience in the subjects covered in lectures.		
Learning Hours: 132 (36L;36Lb;12T;48P)		Prerequisites: PHYS 239/3.0

Course Material

Slides: The primary course material will be Prof. Shastri's slides, posted weekly.

Textbook: The recommended text is *Electronics: A Systems Approach (6th Edition)* by Neil Storey. However, it is *not* necessary to purchase a new one. An older edition would suffice. Any circuit textbook(s) that covers the course topics (listed on pp. 7-8) would suffice.

The text can be purchased through the campus bookstore:

<https://www.campusbookstore.com/textbooks/search-engine/results?Course=PHYS24001>

<https://www.campusbookstore.com/textbooks/search-engine/results?Course=ENPH24002>

Companion website: We recommend students refer to the Companion Website for Storey: Electronics, Sixth Edition. It is available for free.

To view the *video library*:

https://wps.pearsoned.co.uk/ema_uk_he_storey_electronics_6/253/64880/16609466.cw/index.html

To practice *multiple choice questions*:

https://wps.pearsoned.co.uk/ema_uk_he_storey_electronics_6/253/64879/16609133.cw/index.html

Legacy notes: The previous instructor, Prof. Jordan Morelli, created two-part course notes on analog and digital electronics. These notes are posted on the OnQ course webpage. Students are welcome to refer to these notes (but the course will mainly follow Prof. Shastri's slides).

Graduate Attribute Indicators

ENPH 334 develops the Canadian Engineering Accreditation Board Graduate Attributes through the following indicators:

Indicator	Description
KB-NS (Electricity and Magnetism)	Interpret natural phenomena and relationships using analytical and/or experimental techniques.
PA-Solve	Implement solutions for complex engineering problems.
IN-Analysis	Analyze and interpret data using appropriate techniques and tools.
DE-Strategies	Compare multiple strategies for solving a problem.

DE-Solutions	Create a product, process, or system to solve a problem, that meets specified needs, and subject to appropriate iterations.
ET-Apply	Apply and manage appropriate techniques, apparatus, databases, models, tools, and/or processes to accomplish a task.
TW-Contribution	Take initiative to plan, organize and complete tasks, as an individual and team member, to meet goals.
CO-Spoken	Deliver formal and informal oral presentations with suitable language, content, style, timing, and flow while adapting format, content and tone to audience and purpose.
CO-Graphics	Create figures, maps, tables, and drawings to engineering report standards.

Course Learning Outcomes (CLO)

CLO	Description	Indicator
CLO 01	Have a thorough understanding of current, voltage, resistance, and Ohm's law.	KB-NS (Electricity and Magnetism)
CLO 02	Analyze DC circuits using Kirchhoff's laws.	KB-NS (Electricity and Magnetism) IN-Analysis
CLO 03	Analyse AC Circuits using complex representations Impedance, RC filters, decibels, tuned filters.	KB-NS (Electricity and Magnetism) IN-Analysis
CLO 04	Use an Arduino microcontroller to replicate the functionality of a multimeter and an oscilloscope, and to implement functional circuit designs.	ET-Apply
CLO 05	Have a working knowledge of the Orcad circuit simulation software package.	ET-Apply
CLO 06	Be knowledgeable of voltage and current sources, Thevenin's and Norton's theorems.	KB-NS (Electricity and Magnetism)
CLO 07	Analyse circuits containing operational amplifiers, and design complex circuits using operational amplifiers as a building block.	PA-Solve IN-Analysis
CLO 08	Analyse digital electronics: logic gates, flip-flops, counters, and memory with the aid of Truth tables, DeMorgan's theorems, and Boolean algebra.	PA-Solve IN-Analysis
CLO 09	Create figures and diagrams for the analog and digital design project reports.	CO-Graphics
CLO 10	Use multiple strategies to solve an engineering problem as part of design projects.	DE-Strategies DE-Solutions
CLO 11	Deliver short presentations on the design projects by speaking clearly and confidently and answering questions.	CO-Spoken
CLO 12	Works in teams of two to complete labs and design projects.	TW-Contribution

Grading Summary and Marking Scheme

Final Exam	40%
Quizzes (×4) best 3/4	21%
Design Projects (×2)	24%
Laboratory Test	8%
Laboratory Notebook	7%
Total	100%

Quizzes

There will be **four quizzes** during the term, administered during the scheduled tutorial sessions according to the course timetable. Together, the quizzes will account for **21% of your final course grade**. Each quiz will be weighted equally, and your overall quiz grade will be calculated using your **best three of the four quiz marks**.

Quizzes are **open-book and open-notes**. Laptops, phones, tablets, and other electronic devices are not permitted.

Laboratory Work

Labs will start on Sep. 14, 2024. Lab work will be done in **teams of two**. You are welcome to choose your lab partners, but you must **maintain this partnership throughout the term**. Please register your team with the TA in the first lab. Also, we encourage PHYS and APPLE MATH students to partner with ENPH students.

Lab Attendance: Attendance is mandatory, and it will be taken. Students absent or who do not complete labs must make up the labs during one of the other lab sessions. Make the necessary appointments with the TAs.

Completion of Lab Work: All labs must be completed. The evidence of completion will be:

- a) a sign-off or an initial in the lab by the TAs; and
- b) a corresponding check-off in the attendance register.

If *all* labs are not completed satisfactorily, then a zero mark will apply to the entire lab section of the course (39%).

Design Projects

Two laboratory design projects, one **analog** (10% of your course grade) and one **digital** (14% of your course grade), will be **assigned in weeks 6 and 9** and **due in weeks 8 and 11**, respectively. The projects will involve research on topics covered in the lectures.

On the project due dates, students will be allowed 20 minutes of lab time to test and troubleshoot their designs, after which the TAs will evaluate the project.

Students must also submit a written report (with the course cover sheet). The report should contain a brief description of the circuit, circuit diagram, transfer function (i.e., an equation relating output to input), calculations for circuit components, a summary of the purpose and functionality of the design, and references.

The project will be evaluated based on:

- Report handed in
- Presentation in the lab and discussions
- Circuit construction
- Correct circuit operation
- The project marking scheme will be posted on the course OnQ page.
- Note: A standard Title Page (cover sheet) for ENPH334/PHYS334 must be used for reports and signed in the appropriate place.

Laboratory Test

This will be a 20-minute, hands-on, **individual exercise** in the last week of the term. During the test, you must demonstrate that you are familiar with the lab equipment and can make basic electronic measurements successfully. More details of this test will be given closer to the end of the term.

The laboratory test is worth **8% of your overall grade** in the course.

Lab (Electronic) Notebook

A lab notebook records procedures, observations, and your thought process that would enable another scientist to reproduce your observations. It consists of (i) objective information (apparatus sketches, circuit diagrams, instrument settings, data filenames, simple plots, conditions of the experiment; (ii) subjective information: the researcher's interpretation or evaluation of the events in the lab and commonly accompanies the objective information from the experiment; (iii) analysis: preliminary analysis performed on raw data throughout the entire experimental process; and (iv) planning information: this consists of plans or directions for the experiment/project. Keeping track of new ideas and experimental directions can be difficult unless they are written alongside other pertinent information.

Each person must have an electronic notebook to record the measurements made during the lab. An electronic notebook (e.g., Google Docs or OneNote with Jupyter Notebooks) must be used to be easily shared. If you use a physical notebook, it must be scanned as a PDF (to facilitate submission). On the front of your notebook, indicate your name, lab section, and group. Pages in your lab notebook should be numbered, and there should be a table of contents at the beginning of the ENPH 334 or PHYS 334 section (if you use a lab book, you have previously used it in another course).

In your lab book, you should record your notes and calculations when preparing before the lab. During the lab, you should record all information about what you have done so that you, or someone else, can perform the experiment again (hopefully with the same results) in 6 months or a year with only your lab book as a guide. There is no need to copy instructions from the lab instruction sheets into your lab book if they are permanently fastened into your lab book.

The information in your lab book should be complete enough that you can write a formal report of your experiment using only the information it contains. If you use information from the data books available in the lab, the web, or any other source, you should record a reference to your source in your lab book.

Some essential things in your lab notebook:

- Numbered pages and a Table of Contents
- Clear circuit diagrams with component values should be prepared before the lab. The diagrams should be drawn before wiring the circuit and include the device pin numbers.
- Labeled waveform sketches, including scales.

- Tables of data with descriptive titles indicating the table contents. Table formats can be prepared beforehand.
- Graphical presentations of the data recorded during the lab. The graphs must have titles and properly labeled axes. Draw these graphs as you make measurements on a particular circuit. The initial results you have graphed will often guide you to the subsequent measurements you should make and save you considerable time in the lab.
- Summaries and comments on the operation of each circuit. Sufficient circuit analysis should be included (this can also be done outside of the lab time) to allow a comparison of your data with the expected behaviour of the circuit and a discussion of possible causes for any discrepancies. However, a detailed discussion of errors is not required in your lab book. In general, agreement within 5-10% is expected.

Evaluation of Lab Notebook: At some time near the end of the term (during week 11), one of the TAs will request to see your lab book. They will check to ensure they can reproduce what you did in the lab with no information other than what is in your book. There is no need to copy the procedure into your book if you permanently fasten the instruction sheets. You should explicitly note any variations from or changes to the lab instruction sheets.) They will be checking for completeness: table of contents, numbered pages, circuit diagrams, labeled and titled graphs, and a summary of the lab. The laboratory notebook mark is worth **7% of your overall grade** in this course.

The laboratory notebooks will be marked during Week 11 in the lab, but they may be requested earlier at the instructor's or TAs discretion. However, during Week 4, you will submit your Lab Notebooks for feedback only.

Laboratory Equipment

The laboratory is built around laboratory stations in Stirling Hall, which contain the following set of instruments:

- A “bread-board” unit for circuit construction. It includes three power supplies: +15 V at 0.5 A, -15 V at 0.5 A (or 1 A), and +5 V at 1 A (Proto-board 203A or Project-board PP272 in Stirling; Global Specialities Model PB503 in the ILC.)
- A function generator (HP 3311A, Global 2001, or BK Precision 3011 in Stirling; Global Specialities Model 105-2102 in the ILC). This device allows you to generate a variety of waveforms, such as sine waves, square waves, or pulses of the desired amplitude at any given frequency.
- A digital multimeter (HP 3476A or Data Precision 1351 in Stirling; AW Sperry Model DM-8700 in the ILC). A multimeter lets you measure AC or DC voltages, currents, and resistance.
- A dual-trace oscilloscope (Hitachi model V-222 or V-252, both 20 MHz, in Stirling; Tektronix Model TDS 210, 60 MHz, in the ILC). The oscilloscope is one of the most valuable tools for troubleshooting electronic circuits. It can display repetitive waveforms and measure AC or DC voltages (or a combination of both) and frequency. Modern digital oscilloscopes also allow storing a sampled version of any waveform, which can be downloaded to a PC for further analysis.
- The laboratory has a supply of components, tools, wires, and sundries. Each bench has a digital multimeter for measuring resistances and a digital capacitance meter for measuring capacitances.
- A set of reference books is kept in the laboratory. If you use information from these data books or the web when completing your lab reports, you should record the source of your information in your lab book. The data books should not be removed from the lab.

Accommodations

Students who require accommodations for completing quizzes and extra time for the lab project or the lab test must have a note/form from Queen's Health Services or Queen's Counseling Services stating the accommodations required. The Instructor may also grant accommodations in exceptional circumstances such as illness, Court proceedings, or other emergencies. Please do not hesitate to contact the Instructor with any concerns about the course.

Course Schedule

	Monday	Tuesday	Wednesday	Thursday	Friday
08:30 - 09:30	Lab 005 STIRLG 404/405 Fraser/Ammar		Lab 004 STIRLG 404/405 Fraser/Ammar		
09:30 - 10:30					
10:30 - 11:30					
11:30 - 12:30		Tutorial/ Quizzes MACDON 1		Lab 003 STIRLG 404/405 Sam/Cameron	
12:30 - 13:30					
13:30 - 14:30					
14:30 - 15:30		Lecture KINGST 101			
15:30 - 16:30					Lecture KINGST 101
16:30 - 17:30			Lecture KINGST 101		
18:30 - 19:30					
19:30 - 20:30			Lab 002 STIRLG 404/405 Sam/Cameron		
20:30 - 21:30					

Lectures: KINGST 101; 3 hours per week.

For every lecture hour, students should plan to spend an additional hour reinforcing key concepts.

Tutorials: MACDON 1; 1 hour per week.

Laboratories: Stirling 404/405; 3 hours per week.

Total: Lectures 3 hours/week + Tutorials 1 hour/week + Labs 3 hours/week + Preparation 3 hours/week
= **10 hours/week**

Course Timetable

The schedule of topics, as outlined below, may change slightly, but we will try to adhere to it.

Week	Lectures/Theory	Labs	Comments
1 Sep. 7	Review of Basics; Resistors; KCL and KVL; Thévenin's Theorem; Capacitors; Phasors; Complex AC Circuits; Lab Equipment;	No labs	Tutorial will cover lab overview by Fraser McCauley (TA)

2 Sep. 14	Review Phasors; R-C Circuits; Diodes; deciBel (dB); Bode Plots; Generalized Amplifiers	A0 - Simulation Intro	Tutorial will cover lecture material
3 Sep. 21	Op-Amps; Op-Amp Circuits	A1 - Resistors, Capacitors, Diodes	Tutorial will cover lecture material
4 Sep. 28	Op-Amp Parameters (slew rate, rise time, and bandwidth); Noise, and EMC.	A2 - Inverting Amplifier and Voltage Follower	<u>Quiz 1 (Sep. 29)</u> during the tutorial MACDON1 Submit Lab Notebooks (for feedback only)
5 Oct. 5	Comparators; 555 Timer; Feedback basics; Differential Amplifiers; Input and Output Impedance	A3 - Summing Amplifiers (hardware) and The 555 Timer (simulation)	
Thanksgiving and Fall Term Break Oct 12–16			
6 Oct. 19	Effects of Negative Feedback; Active Filters and Their Response	A4 - Simple Comparator and Window Comparator	<u>Quiz 2 (Oct. 20)</u> during the tutorial MACDON1 Analog Lab Projects assigned
7 Oct. 26	Number Systems; Binary Arithmetic; Digital Gates; Switch Debouncing; Multivibrators	Analog Project work	Tutorial to troubleshoot design projects
8 Nov. 2	Boolean Algebra; TTL and CMOS Characteristics; Adders; Latches; Flip-Flops	Analog Project Presentation	Tutorial will go over practice problems
9 Nov. 9	Flip-Flops; D/A Conversion; Counters; Counter Decoding	D1 - Gates & Latches, NAND, AND, Full Adder (simulation), Clocked RS and JK Flip-Flops.	<u>Quiz 3 (Nov. 10)</u> during tutorial MACDON1 Digital Lab Projects Assigned
10 Nov. 16	Ripple Counters; Modulo-N; Up/Down; Synchronous; Shift Registers; Serial-Parallel; Parallel Load Shift Register	D2 - Ripple Counter and D/A Conversion (hardware)	Tutorial to troubleshoot design projects
11 Nov. 23	A/D Conversion - Terminology; Circuits; Methods	<u>Digital Project Presentation</u>	<u>Quiz 4 (Dec. 1)</u> during the tutorial MACDON1 Lab Notebooks Marked
12 Nov. 30	Memory, Terms and Acronyms, Digital Memory, PC Memory	<u>Lab Test</u>	Tutorial will go over practice problems

Notes:

- 1) Electronic Circuit Simulation is part of the lab course for ENPH/PHYS 334 students, using the LTSpice (strongly preferred) or OrCAD software with PSpice. Apart from the introductory labs in Week 2, where the labs will be all simulation, the other simulation labs will have only part of the lab done with LTSpice (or OrCAD). Therefore, the **lab test will include circuit simulation questions**.
- 2) LTSpice, which is available for free at <https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html>

- 3) OrCAD is installed on all the lab workstations. Free OrCAD software for students can be downloaded from <https://www.orcad.com/orcad-academic-program>. OrCAD is only available on Windows. If you are a MAC user, create a dual boot system to install OrCAD or access it on the lab workstations.
- 4) It is expected that 334 students are familiar with the lab equipment in STI 404/405. If not, download and go through the 333 Lab A0 (Lab Equipment).

Statement on Academic Integrity

Academic integrity comprises the five core values of honesty, trust, fairness, respect, and responsibility (see www.academicintegrity.org). These values are central to building, nurturing, and sustaining an academic community in which all community members will thrive. Adherence to the values expressed through academic integrity forms a foundation for the “freedom of inquiry and exchange of ideas” essential to the intellectual life of the University (see the Senate Report on Principles and Priorities: <http://www.queensu.ca/secretariat/policies/senateandtrustees/principlespriorities.html>).

Students are responsible for familiarising themselves with the regulations concerning academic integrity and ensuring that their assignments conform to the principles of academic integrity. Information on academic integrity is available on the Arts and Science website:

<https://www.queensu.ca/artsci/undergrad-students/student-services/academic-integrity>

Departures from academic integrity include plagiarism, use of unauthorized materials, facilitation, forgery, and falsification, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions that contravene the regulation on academic integrity carry sanctions ranging from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the university.

Copyright of Course Materials

The material on this website is copyrighted and is for the sole use of students registered in the ENPH 334 or PHYS 334 course. The material on this website may be downloaded for a registered student's personal use, but it shall not be distributed or disseminated to anyone other than students registered in the courses noted above. Failure to abide by these conditions is a breach of copyright and may also constitute a breach of academic integrity under the University Senate's Academic Integrity Policy Statement.

Disability Accommodation Statement

Queen's University is committed to achieving full accessibility for persons with disabilities. Part of this commitment includes arranging academic accommodations for students with disabilities to ensure they have an equitable opportunity to participate in all their academic activities.

If you are a student with a disability and need accommodations, you are strongly encouraged to contact Student Wellness Services (SWS) and register as early as possible. For more information, including important deadlines, please visit the Student Wellness website at:

<http://www.queensu.ca/studentwellness/accessibility-services/>

Calculator Policy

The only calculator permitted for the ENPH 334 and PHYS 334 courses is the non-communicating Casio 991 series calculator (see Policy excerpts below for an explanation).

Arts & Science: Calculators acceptable during quizzes, tests, and examinations are intended to support the basic calculating functions required by most Arts and Science courses. For this purpose, the Casio 991 series calculator is permitted and is the only approved calculator for Arts and Science students. This inexpensive calculator is sold on Amazon, Staples, and other popular suppliers of school and office supplies.

See: <https://www.queensu.ca/academic-calendar/nursing/bnsc-academic-regulations/a-r-13-use-of-calculators/>

Engineering – Policy on using calculators for examinations: The Casio 991 will be the only calculator approved for engineering exams.

See: <http://my.engineering.queensu.ca/policy/Calculator.html>

Calculators are divided into three classes:

- a) Communicating Calculators: These are never permitted in examinations.
- b) Non-Communicating Calculators with text storage and/or graphing capability and/or long-term memory: Instructors may permit the use of these calculators in examinations but only if students are also permitted to bring “significant amounts” of written material to offset the advantage that otherwise accrues to those with fancy calculators. Such a calculator must carry a “red sticker.”
- c) Non-communicating calculators without the features mentioned above: Instructors may permit the use of such calculators in any examination. To be used in an examination, such a calculator must carry a “gold or blue sticker,” except the prevalent Casio 991 model, which may be used without a sticker.